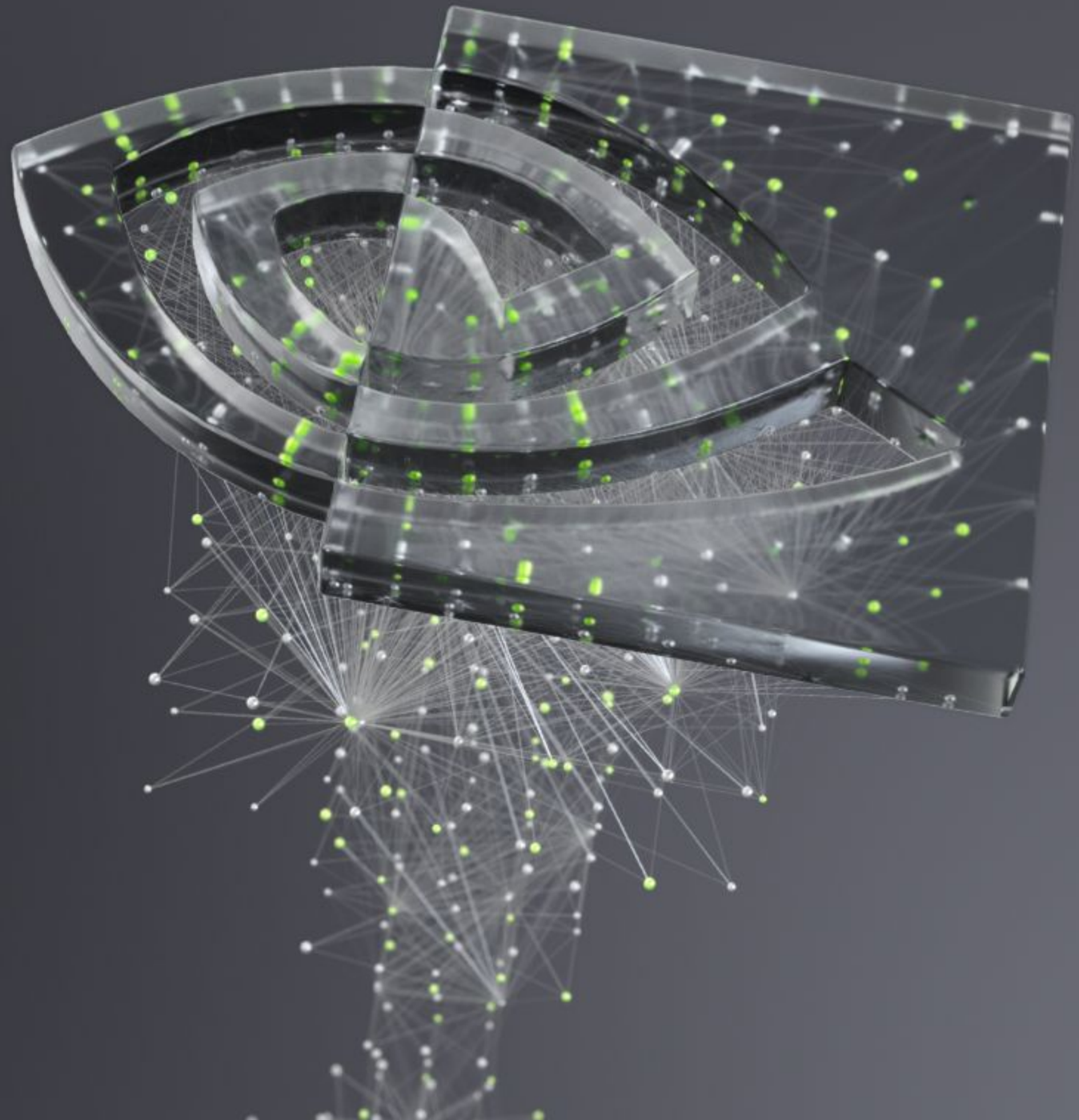


DENT

Roopa Prabhu

Aug 14, 2020

The World of Open Network Operating systems

- Cumulus Linux - Debian based distribution, standardizes on Linux API (Linux API to vendor SDK API abstraction) [1]
- SONIC - standardizes the user-space SDK API (SAI abstraction) [2]
- DANOS - Autonomous NOS which started with seed code from AT&T [3]
- Mellanox Linux switch - any linux distribution + mlxsw driver [4]
- ONL - Not a NOS but provides a base and platform abstraction for an Open NOS (many other NOSes are based on ONL) [5]
- and more ..

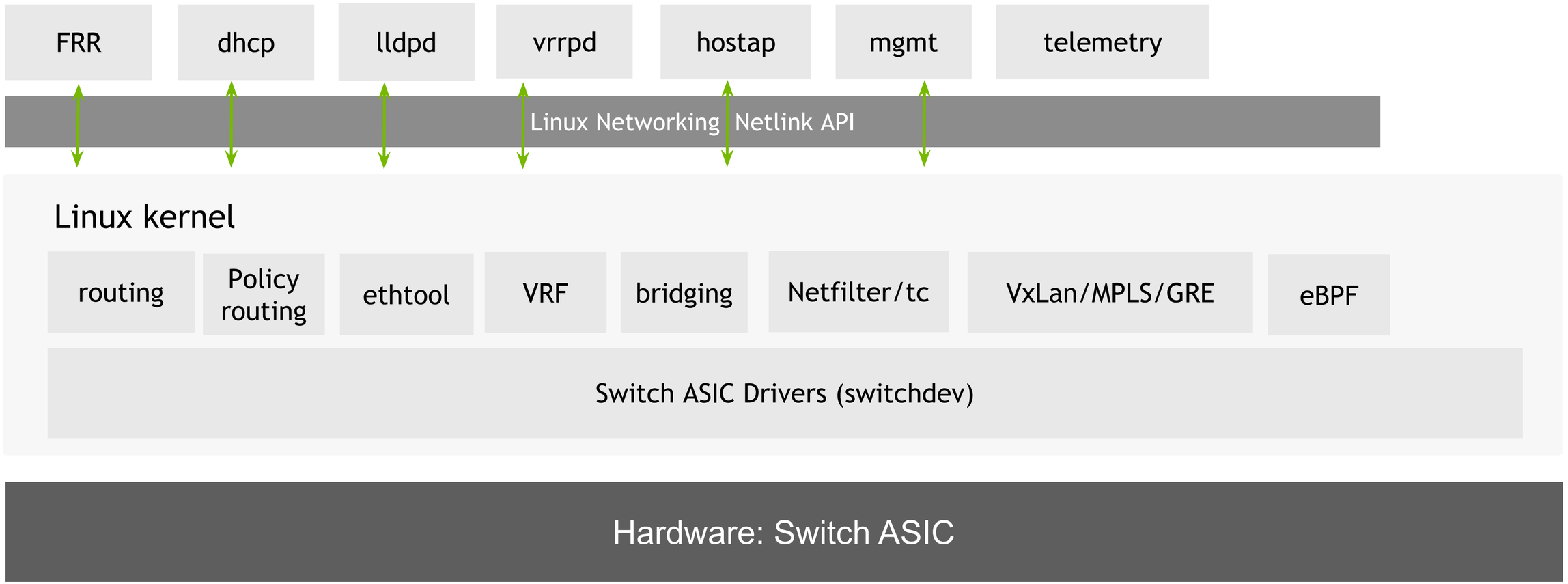
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DENT NOS

- New NOS for Distributed Enterprise [6]
  - Network Edge and Campus
  - Simplifying Retail/Campus
  - Cost-effective
- Founding members: Amazon, Mellanox, Cumulus, Delta, Marvell, WNC
- Standardizes on Linux API and switchdev for offload to switch ASIC

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Architecture



DENT hardware

- Wistron Neweb Corp (WNC) Box - Marvell ASIC
- Delta Box - Mellanox Spectrum ASIC
- more coming ...

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DENT OS components

- ONL - Debian base, provides a hardware platform abstraction (ONLP [7])
- Switchdev drivers
- Linux platform drivers or ONLP drivers and API
- Software and control plane: debian ecosystem, lldpd, FRR, ifupdown2
- Additional capabilities: power-over-ethernet

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Linux Platform drivers

- Platform drivers to manage: fan, psu, thermal, sfp, leds
  - sysfs - enumeration of devices and buses
  - udev
  - Linux utilities:
    - pwmconfg
    - fancontrol
    - sensors
    - ethtool

  [example mellanox implementation and documentation: [8]]

- Future: ACPI ? one such effort was done by Cumulus under OCP project - APD [9]

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Switch ASIC offload drivers

- Mellanox switchdev in-tree mlxsw [10]
- Marvell prestera switchdev driver (Upstream inclusion WIP) [11]

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Where to find DENT

Mailing List
https://lists.dent.dev/g/tsc

Seed code waiting to be pushed:
https://github.com/dentproject

Talk at ONES:
https://events.linuxfoundation.org/open-networking-edge-summit-north-america/

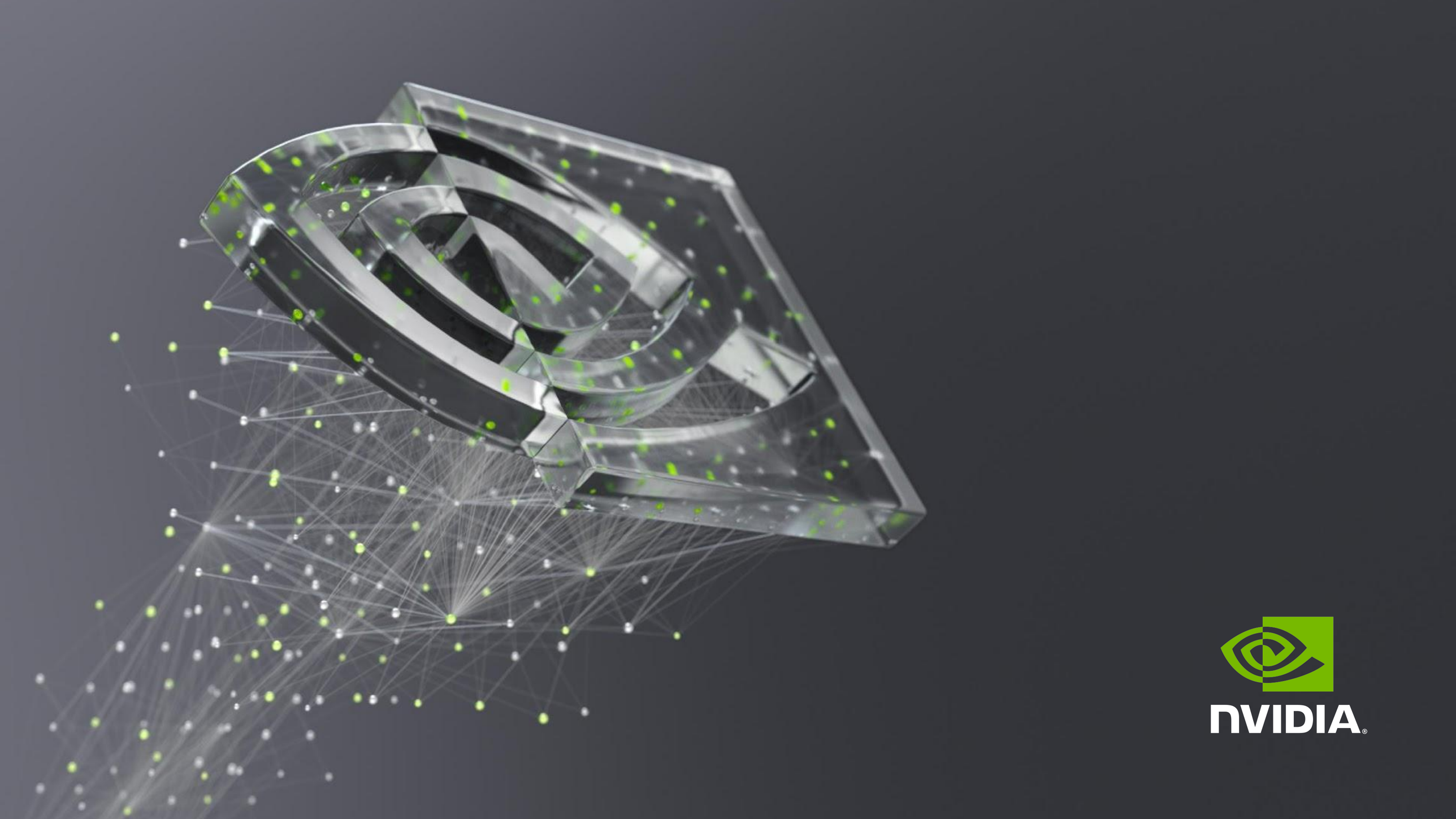
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https://git.kernel.org/pub/scm/linux/kernel/git/netdev/net-next.git/tree/drivers/net/ethernet/mellanox/mlxsw
[11] Marvell prestera switchdev driver:

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